Final Lab Project — Instructions

Mobile Application and Development Lab (0613-410) ·

1. Overview

- **Individual project**.
- **Lab time:** From Oct 25, 2026, all lab classes are for project work. If you have trouble running Android Studio on your own computer, you can work on the lab PCs. Working on your own laptop will earn bonus marks. I will check your project progress in every lab class.
- **What to build:** one Android app that solves a real problem.
- **Tool:** Android Studio (Java + XML), with Firebase for login and the database
- **Submit:** the app (APK), the GitHub repository, and a lab report.

2. Features your app must have

All 12 must work and be used for a real purpose in the app.

#	Feature	How to do it in Android Studio
1	At least 5 screens with navigation (Splash, Login, Register, Home, Add/Edit)	Activities + `Intent`, pass data with `putExtra() if needed`
2	Layouts and basic views (text, image, button, card)	`ConstraintLayout` / `LinearLayout`, `TextView`, `ImageView`, `Button`, `CardView`
3	User input with validation (no crash on empty/wrong input)	`TextInputEditText` + `setError()`
4	Model class for your data	Java class with getters/setters
5	Scrolling list of records	`RecyclerView` + Adapter + ViewHolder
6	Add, view, edit, delete (CRUD) with dialog and FAB	`AlertDialog`, `FloatingActionButton`

#	Feature	How to do it in Android Studio
7	Firebase login: register, login, logout, stay logged in	Tools → Firebase → Authentication; `FirebaseAuth`
8	Data saved in Firebase database, list updates in real time	Firebase Realtime Database or Cloud Firestore + a data change listener
9	Save a setting on the phone (e.g. remember me)	`SharedPreferences`
10	Take or pick a photo and show it	Camera / Gallery `Intent`
11	Use one sensor (e.g. shake, proximity)	`SensorManager` + `SensorEventListener`
12	Open another app (call, share, web link)	Implicit `Intent` (`ACTION_DIAL` , `ACTION_SEND` , `ACTION_VIEW`)

Also: custom app name and icon, Material Design theme, and SnackBar/Toast messages for success or error.

Bonus (extra marks): Google Maps, REST API (weather etc.), SQLite/Room, navigation drawer or tabs, Google Sign-In.

3. GitHub

1. Create a repository and Keep it **public**.
2. Connect your project to it from Android Studio and push from the first day.
3. Commit regularly with clear messages.
4. Add a `README.md` : app name, your name and ID, feature list, and steps to run the app.
5. Create a GitHub **Release** `v1.0` and attach your APK.
6. Do not delete or make the repository private after submission. You can keep improving the app and push updates later.

4. Lab report

PDF, 10–15 pages. Include these sections:

1. Cover page (course, title, your name and ID)
2. Introduction: the problem and your app's goal

3. Features with screenshots
4. **Components used, alternatives, and why you chose them** (main part of the report)
5. Firebase database structure
6. GitHub link
7. Problems faced and how you solved them
8. Conclusion and references

For item 4 (Components used), list **every component you actually used** in your app. For each one, write what it does in your app, name its alternatives, and explain in 2–3 lines why you chose it **for your app**.

The table below is only a **sample**. Your list will be different: add the components you used and leave out the ones you did not use.

Component	Alternatives
`ConstraintLayout`	`LinearLayout`, `RelativeLayout`
`RecyclerView`	`ListView` + `ArrayAdapter`, `GridView`
Activities + Intents	Fragments + Navigation Component
Firebase Authentication	Google Sign-In, phone OTP, own server login
Firebase Realtime Database	SQLite, Room
SharedPreferences	SQLite, Room, text file
Camera / Gallery `Intent`	CameraX
`AlertDialog`	Bottom sheet, separate Activity
Git + GitHub	GitLab, Bitbucket

5. Submission and marks

- [ ] APK attached to GitHub Release `v1.0`
- [ ] Repository link and report PDF submitted to the instructor
- [ ] Live demo and viva in the final lab

Part	Marks
Required features working	40
UI and code quality	15
GitHub use	15
Lab report	15
Demo and viva	15

Use of AI tools: You may use the Gemini agent in Android Studio. However, you must review and modify all the code it generates. You must understand every feature in your app and how it works, because you will have to explain each one in the viva.